

1582. Special Positions in a Binary Matrix

Solved

Easy

Topics

Companies

Hint

Given an $m \times n$ binary matrix `mat`, return *the number of special positions in* `mat`.

A position (i, j) is called **special** if `mat[i][j] == 1` and all other elements in row `i` and column `j` are `0` (rows and columns are **0-indexed**).

Example 1:

1	0	0
0	0	1
1	0	0

Input: `mat = [[1,0,0],[0,0,1],[1,0,0]]`

Output: 1

Explanation: $(1, 2)$ is a special position because `mat[1][2] == 1` and all other elements in row 1 and column 2 are 0.

Example 2:

1	0	0
0	1	0
0	0	1

Input: `mat = [[1,0,0],[0,1,0],[0,0,1]]`

Output: 3

Explanation: (0, 0), (1, 1) and (2, 2) are special positions.

Constraints:

- `m == mat.length`
- `n == mat[i].length`
- `1 <= m, n <= 100`
- `mat[i][j]` is either 0 or 1.

Python:

class Solution:

```
def numSpecial(self, mat):  
    m, n = len(mat), len(mat[0])  
    row = [0] * m  
    col = [0] * n
```

```
    for i in range(m):
```

```

        for j in range(n):
            if mat[i][j] == 1:
                row[i] += 1
                col[j] += 1

    return sum(
        mat[i][j] == 1 and row[i] == 1 and col[j] == 1
        for i in range(m)
        for j in range(n)
    )

```

JavaScript:

```

var numSpecial = function(mat) {
    const m = mat.length;
    const n = mat[0].length;

    const row = new Array(m).fill(0);
    const col = new Array(n).fill(0);
    for (let i = 0; i < m; i++) {
        for (let j = 0; j < n; j++) {
            if (mat[i][j] === 1) {
                row[i]++;
                col[j]++;
            }
        }
    }

    let ans = 0;
    for (let i = 0; i < m; i++) {
        for (let j = 0; j < n; j++) {
            if (mat[i][j] === 1 && row[i] === 1 && col[j] === 1) {
                ans++;
            }
        }
    }

    return ans;
};

```

Java:

```

class Solution {
    public int numSpecial(int[][] mat) {
        int m = mat.length, n = mat[0].length;
        int[] row = new int[m];
        int[] col = new int[n];
    }
}

```

```
for (int i = 0; i < m; i++) {
    for (int j = 0; j < n; j++) {
        if (mat[i][j] == 1) {
            row[i]++;
            col[j]++;
        }
    }
}

int ans = 0;
for (int i = 0; i < m; i++) {
    for (int j = 0; j < n; j++) {
        if (mat[i][j] == 1 && row[i] == 1 && col[j] == 1) {
            ans++;
        }
    }
}

return ans;
}
```